

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

- Let $f(x) = [\tan^2 x]$, (where $[.]$ denotes greatest integer function). Then -
 (A) $\lim_{x \rightarrow 0} f(x)$ does not exist (B) $f(x)$ is continuous at $x = 0$.
 (C) $f(x)$ is not differentiable at $x = 0$ (D) $f'(0) = 1$
MDY001
- Let f be differentiable at $x = 0$ and $f'(0) = 1$. Then $\lim_{h \rightarrow 0} \frac{f(h) - f(-2h)}{h} =$
 (A) 3 (B) 2 (C) 1 (D) -1
MDY002
- Let $g(x) = \begin{cases} 3x^2 - 4\sqrt{x} + 1 & \text{for } x < 1 \\ ax + b & \text{for } x \geq 1 \end{cases}$.
 If $g(x)$ is continuous and differentiable at $x = 1$ then -
 (A) $a = b = 4$ (B) $a = b = -4$
 (C) $a = 4$ and $b = -4$ (D) $a = -4$ and $b = 4$
MDY003
- The function $g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \geq 0 \end{cases}$ can be made differentiable at $x = 0$ -
 (A) if b is equal to zero (B) if b is not equal to zero
 (C) if b takes any real value (D) for no value of b
MDY004
- Which one of the following functions is continuous everywhere in its domain but has atleast one point where it is not differentiable?
 (A) $f(x) = x^{1/3}$ (B) $f(x) = \frac{|x|}{x}$ (C) $f(x) = e^{-x}$ (D) $f(x) = \tan x$
MDY005
- If $f(x) = (x^5 + 1)|x^2 - 4x - 5| + \sin|x| + \cos(|x - 1|)$, then $f(x)$ is not differentiable at -
 (A) 2 points (B) 3 points (C) 4 points (D) zero points
MDY006
- Let $f(x) = \begin{cases} x^3 + 2x^2 & x \in \mathbb{Q} \\ -x^3 + 2x^2 + ax & x \notin \mathbb{Q} \end{cases}$, then the integral value of ' a ' so that $f(x)$ is differentiable at $x = 1$, is
 (A) 2 (B) 6 (C) 7 (D) not possible
MDY007
- Let $\mathbb{R}$ be the set of real numbers and $f : \mathbb{R} \rightarrow \mathbb{R}$, be a differentiable function such that $|f(x) - f(y)| \leq |x - y|^3 \forall x, y \in \mathbb{R}$. If $f(10) = 100$, then the value of $f(20)$ is equal to -
 (A) 0 (B) 10 (C) 20 (D) 100
MDY008

9. The function $f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$ is NOT differentiable at :
 (A) -1 (B) 0 (C) 1 (D) 2

MDY009

10. If $f(x) = \begin{cases} e^{x^2} + x^3 + 1 & ; x > 0 \\ ax^2 + bx + 2 & ; x \leq 0 \end{cases}$ is differentiable at $x = 0$, then -
 (A) $a \in R$ & $b = 0$ (B) $a \in R$ & $b = 1$ (C) $a \in R$ & $b = 2$ (D) $a \in R$ & $b = 3$

MDY010

11. If $f(x) = [\sin x] + \sqrt{\sin x - [\sin x]}$, where $[.]$ denotes greatest integer function, then $f(x)$ is -
 (A) differentiable at $x = \frac{\pi}{2}$ & $x = \pi$ both
 (B) differentiable at $x = \frac{\pi}{2}$ but not at $x = \pi$
 (C) differentiable at $x = \pi$ but not at $x = \frac{\pi}{2}$
 (D) neither differentiable at $x = \pi$ nor at $x = \frac{\pi}{2}$

MDY011

12. If $f(x) = \begin{cases} \sin^{-1} x + a \cos \pi x & -1 < x < 0 \\ b \cos^{-1} x + \sin \pi x & 0 \leq x < 1 \end{cases}$ is differentiable in $(-1, 1)$, then $(2a + b)$ is equal to -
 (A) π^2 (B) $1 + \pi^2$ (C) $1 - \pi^2$ (D) $\pi^2 - 1$

MDY012

13. Number of points where $f(x) = |x - \operatorname{sgn}(x)|$ is non-differentiable, is ($\operatorname{sgn}(\cdot)$ denotes signum function)
 (A) 0 (B) 1 (C) 2 (D) 3

MDY152

14. Let $f(x) = \begin{cases} \frac{\ln \cos x}{ax} & , x > 0 \\ 0 & , x = 0 \\ \frac{e^{x^2} - 1}{bx} & , x < 0 \end{cases}$. If $f'(0) = \frac{1}{4}$, then
 (A) $a + b = 2$ (B) $b - a = 2$ (C) $a + b = -2$ (D) $b - a = 4$

MDY153

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

- If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then -

(A) $Rf'(0)$ exist (B) $Lf'(0)$ exist but $Rf'(0)$ does not exist
(C) $\lim_{x \rightarrow 0^+} f(x)$ exist (D) $f(x)$ is differentiable at $x = 0$

MDY041
- The function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \left(\frac{x^2}{4}\right) - \left(\frac{3x}{2}\right) + \left(\frac{13}{4}\right), & x < 1 \end{cases}$ is -

(A) continuous at $x = 1$ (B) differentiable at $x = 1$
(C) continuous at $x = 3$ (D) differentiable at $x = 3$

MDY042
- Select the correct statements -

(A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \leq 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.
(B) The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.
(C) If f is continuous at $x = 5$ and $f(5) = 2$ then $\lim_{x \rightarrow 2} f(4x^2 - 11)$ exists
(D) If $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ and $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$ then $\lim_{x \rightarrow a} f(x) \cdot g(x)$ need not exist.

MDY043
- If $f(x) = \text{sgn}(x^5)$, then which of the following is/are **false** (where sgn denotes signum function) -

(A) $f'(0^+) = 1$
(B) $f'(0^-) = -1$
(C) f is continuous but not differentiable at $x = 0$
(D) f is discontinuous at $x = 0$

MDY044
- Graph of $f(x)$ is shown in adjacent figure, then

(A) $f(x)$ has non-removable discontinuity at two points
(B) $f(x)$ is non-differentiable at three points in its domain
(C) $\lim_{x \rightarrow 1} f(f(x)) = 1$
(D) Number of points of discontinuity = number of points of non-differentiability

MDY045
- Let S denotes the set of all points where $\sqrt[5]{x^2|x|^3} - \sqrt[3]{x^2|x|} - 1$ is not differentiable then S is a subset of -

(A) $\{0, 1\}$ (B) $\{0, 1, -1\}$ (C) $\{0, 1\}$ (D) $\{0\}$

MDY046

7. Which of the following statements is/are correct ?
 (A) There exists a function $f : [0,1] \rightarrow \mathbb{R}$ which is discontinuous at every point in $[0,1]$ & $|f(x)|$ is continuous at every point in $[0,1]$
 (B) Let $F(x) = f(x) \cdot g(x)$. If $f(x)$ is differentiable at $x = a$, $f(a) = 0$ and $g(x)$ is continuous at $x = a$ then $F(x)$ is always differentiable at $x = a$.
 (C) If $Rf'(a) = 2$ & $Lf'(a) = 3$, then $f(x)$ is non-differentiable at $x = a$ but will be always continuous at $x = a$
 (D) If $f(a)$ and $f(b)$ possess opposite signs then there must exist at least one solution of the equation $f(x) = 0$ in (a, b) provided f is continuous on $[a, b]$

MDY047

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x . Then which of the following is/are not always true-
 (A) If f is continuous then g is also continuous
 (B) If f is one-one then g is also one-one
 (C) If f is onto then g is also onto
 (D) If f is differentiable then g is also differentiable

MDY048

9. The function $\phi(x) = [x] - \sin|x|$ (where $[.]$ denotes greater integer function) is -
 (A) derivable at $x = 0$
 (B) continuous at $x = 0$
 (C) $\lim_{x \rightarrow 0} \phi(x)$ does not exists
 (D) continuous and derivable at $x = 0$

MDY049

10. Let $f(x) = \begin{cases} x^2 \cos \frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin \frac{1}{x}, & x > 0 \end{cases}$, then which of the following is (are) correct?
 (A) $f(x)$ is continuous but not differentiable at $x = 0$
 (B) $f(x)$ is continuous and differentiable at $x = 0$
 (C) $f'(x)$ is continuous but not differentiable at $x = 0$
 (D) $f'(x)$ is discontinuous at $x = 0$

MDY050

11. $f(x) = \begin{cases} a + \sin(\sin x); & x \geq 0 \\ \ln(\cos x) + bx; & x < 0 \end{cases}$ is differentiable at $x = 0$, then -
 (A) $a = 1$ (B) $b = 0$ (C) $a = 0$ (D) $b = 1$

MDY051

12. Let $f(x+y) + f(x-y) = 2f(x) \quad \forall x, y \in \mathbb{R}$, then which of the following is/are be correct ($f(0) = 0$ & f is differentiable $\forall x \in \mathbb{R}$)
 (A) f must be odd (B) f must be even (C) $f'(1) = f'(2)$ (D) $f'(2) = f'(3) = f'(4)$

MDY052

13. Let $f(x) = \begin{cases} 1-x, & (0 \leq x \leq 1) \\ x+2, & (1 < x < 2) \\ 4-x, & (2 \leq x \leq 4) \end{cases}$. Then in $0 \leq x \leq 4$ function $y = f(f(x))$ is
 (A) continuous at $x = 1$
 (B) not differentiable at $x = 1$
 (C) continuous and differentiable at $x = 2$
 (D) continuous but not differentiable at $x = 3$

MDY053

14. If $\lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos \frac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at $x = 0$, then the value of $|n + 2m|$ is divisible by
 (A) 2 (B) 3 (C) 5 (D) 7

MDY054

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 15 to 16

A new function tignum (x) (i.e. $\text{tgn}(x)$) is defined such that $\text{tgn}(x) = \begin{cases} 1 & \text{if } [x] = \text{even} \\ -1 & \text{if } [x] = \text{odd} \end{cases}$,

where $[.]$ denotes greatest integer function. Let $f(x) = \text{tgn}(x) \sin(x) |x|$.

On the basis of above information, answer the following questions :

15. Number of points in $[0, 10]$ where $f(x)$ is discontinuous is -
 (A) 8 (B) 9 (C) 10 (D) 11
16. Number of points in $[-10, 10]$ where $f(x)$ is non-derivable is -
 (A) 18 (B) 19 (C) 20 (D) 21

MDY055

MDY056

Paragraph for Question No. 17 to 18

Consider the function $f(x) = \begin{cases} x^2 - 5x + 6 & ; -2 \leq x \leq 4 \\ k - \tan\left(\frac{\pi x}{4}\right) & ; 4 < x \leq 5 \\ \log_{10}(\alpha x) & ; 5 < x \leq 10 \end{cases}$. Let $f_1(x) = |f(|x|)|$.

On the basis of above information, answer the following questions :

17. If $f(x)$ is continuous in $[-2, 10]$ then -
 (A) $\alpha = 2$ (B) $\alpha = 4$ (C) $k = 2$ (D) $k = 4$
18. If $f(x)$ is continuous in $[-2, 10]$, then the number of points, where $f_1(x)$ is NOT differentiable, is -
 (A) 9 (B) 8 (C) 7 (D) 5

MDY057

MDY058

MATRIX MATCH TYPE QUESTION

19. **List-I** **List-II**
- (P) If $f(x)$ is derivable at $x = 3$ & $f'(3) = 2$, (1) 0
 then $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ equals
- (Q) Let $f(x)$ be a function satisfying the condition (2) 1
 $f(-x) = f(x)$ for all real x . If $f'(0)$ exists, then its value is equal to
- (R) For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ (3) 2
 the derivative from the left $Lf'(0)$ equals
- (S) The number of points at which the function (4) 3
 $f(x) = \max\{a - x, a + x, b\}$, $-\infty < x < \infty$,
 $0 < a < b$ cannot be differentiable is
- (A) (P) $\rightarrow$ 3, (Q) $\rightarrow$ 2, (R) $\rightarrow$ 1, (S) $\rightarrow$ 3 (B) (P) $\rightarrow$ 4, (Q) $\rightarrow$ 1, (R) $\rightarrow$ 2, (S) $\rightarrow$ 4
 (C) (P) $\rightarrow$ 3, (Q) $\rightarrow$ 1, (R) $\rightarrow$ 2, (S) $\rightarrow$ 3 (D) (P) $\rightarrow$ 2, (Q) $\rightarrow$ 2, (R) $\rightarrow$ 1, (S) $\rightarrow$ 3

MDY059

EXERCISE (S)

1. If the function $f(x)$ defined as $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \leq 0 \\ x^n \sin \frac{1}{x} & \text{for } x > 0 \end{cases}$ is continuous but not derivable at $x = 0$ then sum of all non-negative integral value(s) of 'n' is
MDY013
2. If $f(x) = |x - 1| \cdot ([x] - [-x])$, where $[x]$ denotes greatest integer function, then the value of $(Rf'(1) + Lf'(1))$ is
MDY014
3. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-3, 3]$ by $f(x) = \begin{cases} -x - [-x] & \text{if } [x] \text{ is even} \\ x - [x] & \text{if } [x] \text{ is odd} \end{cases}$
If L denotes the number of point of discontinuity and M denotes the number of points of non-derivability of $f(x)$, then find $(L + M)$.
MDY015
4. A derivable function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the condition $f(x) - f(y) \geq \ln(x/y) + x - y$ for every $x, y \in \mathbb{R}^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$.
MDY016
5. Let $f(x) = [[x] + \{x^2\}] + \{[x^2] + \{x\}\}$, then number of points where $|f(x)|$ is non-derivable in $[-3, 3]$ is equal to (where $[.]$ denotes greatest integer function & $\{.\}$ denotes fractional part function)
MDY017
6. If f is a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y) + f(0)}{3}$, $\forall x, y \in \mathbb{R}$ and $f'(0) = 2$, then the slope of tangent to $f(x)$ at $x = 7$ is
MDY018
7. Let $f(x)$ be a differentiable function such that $\lim_{x \rightarrow 0} \frac{f(3 - \sin x) - f(3 + x)}{x} = 8$, then $|f'(3)|$ is
MDY019
8. Let $f(x)$ be a differentiable function such that $2f(x + y) + f(x - y) = 3f(x) + 3f(y) + 2xy$ $\forall x, y \in \mathbb{R}$ & $f'(0) = 0$, then $f(10) + f'(10)$ is equal to
MDY020
9. Let $[x]$ denote the greatest integer less than or equal to x . The number of integral points in $[-1, 1]$ where $f(x) = [x \sin \pi x]$ is differentiable are
MDY130
10. Find the natural number 'a' for which $\sum_{k=1}^n f(a+k) = 2048(2^n - 1)$, where the function 'f' satisfies the relation $f(x + y) = f(x) \cdot f(y)$ for all natural numbers x & y and further $f(1) = 2$
MDY131

EXERCISE - JEE (Main) PYQ

- Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that $|f(x) - f(y)| \leq 2|x - y|^{\frac{3}{2}}$, for all $x, y \in \mathbb{R}$. If $f(0) = 1$ then $\int_0^1 f^2(x)dx$ is equal to [JEE (Main) 2019]
 (1) 0 (2) $\frac{1}{2}$ (3) 2 (4) 1
MDY022
- Let $f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$
 Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S : [JEE (Main) 2019]
 (1) is an empty set (2) equals $\{-2, -1, 1, 2\}$
 (3) equals $\{-2, -1, 0, 1, 2\}$ (4) equals $\{-2, 2\}$
MDY023
- Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$ and $g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is:- [JEE (Main) 2019]
 (1) differentiable at all points (2) not differentiable at two points
 (3) not continuous (4) not differentiable at one point
MDY027
- Let S be the set of points where the function, $f(x) = |2 - |x - 3||$, $x \in \mathbb{R}$, is not differentiable. Then $\sum_{x \in S} f(f(x))$ is equal to _____. [JEE (Main) 2020]
MDY030
- Suppose a differentiable function $f(x)$ satisfies the identity $f(x + y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3)$ is equal to. [JEE (Main) 2020]
MDY031
- The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \geq 1 \\ \frac{1}{2}(|x| - 1), & |x| < 1 \end{cases}$ is : [JEE (Main) 2020]
 (1) continuous on $\mathbb{R} - \{1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$.
 (2) both continuous and differentiable on $\mathbb{R} - \{-1\}$.
 (3) continuous on $\mathbb{R} - \{-1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$.
 (4) both continuous and differentiable on $\mathbb{R} - \{1\}$
MDY032
- If the function $f(x) = \begin{cases} k_1(x - \pi)^2 - 1, & x \leq \pi \\ k_2 \cos x, & x > \pi \end{cases}$ is twice differentiable, then the ordered pair (k_1, k_2) is equal to : [JEE (Main) 2020]
 (1) $\left(\frac{1}{2}, 1\right)$ (2) $(1, 1)$ (3) $\left(\frac{1}{2}, -1\right)$ (4) $(1, 0)$
MDY033

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x}\right) + 5x^2 & , x < 0 \\ 0 & , x = 0 \\ x^5 \cos\left(\frac{1}{x}\right) + \lambda x^2 & , x > 0 \end{cases}$

The value of λ for which $f''(0)$ exists, is.

[JEE (Main) 2020]

MDY034

9. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{x, x^2\}$. Let S denote the set of all points in $\mathbb{R}$, where f is not differentiable. Then:

[JEE (Main) 2020]

- (1) $\{0, 1\}$ (2) $\{0\}$ (3) $\emptyset$ (an empty set) (4) $\{1\}$

MDY035

10. Let a function $g: [0, 4] \rightarrow \mathbb{R}$ be defined as $g(x) = \begin{cases} \max\{t^3 - 6t^2 + 9t - 3\}, & 0 \leq x \leq 3 \\ 4 - x & , 3 < x \leq 4 \end{cases}$

then the number of points in the interval $(0, 4)$ where $g(x)$ is NOT differentiable, is _____.

[JEE (Main) 2021]

MDY036

11. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \leq 2 \\ 0 & \text{if } |x| > 2 \end{cases}$$

Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(x) = f(x+2) - f(x-2)$. If n and m denote the number of points in $\mathbb{R}$ where g is not continuous and not differentiable, respectively, then $n + m$ is equal to _____.

[JEE (Main) 2021]

MDY037

12. The number of points, at which the function $f(x) = |2x + 1| - 3|x + 2| + |x^2 + x - 2|$, $x \in \mathbb{R}$ is not differentiable, is _____.

[JEE (Main) 2021]

MDY038

13. Let $[t]$ denote the greatest integer less than or equal to t . Let $f(x) = x - [x]$, $g(x) = 1 - x + [x]$, and $h(x) = \min\{f(x), g(x)\}$, $x \in [-2, 2]$. Then h is :

[JEE (Main) 2021]

- (1) continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$
(2) not continuous at exactly three points in $[-2, 2]$
(3) continuous in $[-2, 2]$ but not differentiable at exactly three points in $(-2, 2)$
(4) not continuous at exactly four points in $[-2, 2]$

MDY039

14. The function $f(x) = |x^2 - 2x - 3| \cdot e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly :

[JEE (Main) 2021]

- (1) four points (2) three points (3) two points (4) one point

MDY040

15. If $[t]$ denotes the greatest integer $\leq t$, then number of points, at which the function

$$f(x) = 4|2x + 3| + 9\left[x + \frac{1}{2}\right] - 12[x + 20]$$

is not differentiable in the open interval $(-20, 20)$, is.

[JEE (Main) 2022]

MDY073

16. Let $f(x) = \min \{1, 1 + x \sin x\}$, $0 \leq x \leq 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to

[JEE (Main) 2022]

- (1) (2, 0) (2) (1, 0) (3) (1, 1) (4) (2, 1)

MDY074

17. Let $f: (-2, 2) \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x[x] & , -2 < x < 0 \\ (x-1)[x] & , 0 \leq x < 2 \end{cases}$$

Where $[x]$ denotes the greatest integer function. If m and n respectively are the number of points in $(-2, 2)$ at which $y = |f(x)|$ is not continuous and not differentiable, then $m + n$ is equal to ____.

[JEE (Main) 2023]

MDY075

18. Let $[x]$ denote the greatest integer function and $f(x) = \max \{1 + x + [x], 2 + x, x + 2[x]\}$, $0 \leq x \leq 2$. Let m be the number of points in $[0, 2]$, where f is not continuous and n be the number of points in $(0, 2)$, where f is not differentiable. Then $(m + n)^2 + 2$ is equal to:

[JEE (Main) 2023]

- (1) 11 (2) 2 (3) 6 (4) 3

MDY076

19. Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$; Then at $x = 0$

[JEE (Main) 2023]

- (1) f is continuous but not differentiable
(2) f is continuous but f' is not continuous
(3) f and f' both are continuous
(4) f' is continuous but not differentiable

MDY077

20. Let $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ for all $x, y, f(y) \neq 0$. If $f'(1) = 2024$, then

[JEE (Main) 2024]

- (1) $xf'(x) - 2024f(x) = 0$ (2) $xf'(x) + 2024f(x) = 0$
(3) $xf'(x) + f(x) = 2024$ (4) $xf'(x) - 2023f(x) = 0$

MDY128

21. If the function $f(x) = \begin{cases} \frac{1}{|x|} & , |x| \geq 2 \\ ax^2 + 2b & , |x| < 2 \end{cases}$ is differentiable on $\mathbb{R}$, then $48(a + b)$ is equal to ____.

[JEE (Main) 2024]

MDY129

EXERCISE - JEE (Advanced) PYQ

1. Let $f(x) = \begin{cases} x^2 \cos \frac{\pi}{x} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$, $x \in \mathbb{R}$, then f is - [JEE (Advanced) 2012]

- (A) differentiable both at $x = 0$ and at $x = 2$
 (B) differentiable at $x = 0$ but not differentiable at $x = 2$
 (C) not differentiable at $x = 0$ but differentiable at $x = 2$
 (D) differentiable neither at $x = 0$ nor at $x = 2$

MDY061

2. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : [0, \infty) \rightarrow \mathbb{R}$, $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2; f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

$$\text{and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

List-I

- P. f_4 is
 Q. f_3 is
 R. $f_2 \circ f_1$ is
 S. f_2 is

List-II

1. onto but not one-one
 2. neither continuous nor one-one
 3. differentiable but not one-one
 4. continuous and one-one

[JEE (Advanced) 2014]

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

MDY062

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define

$$h : \mathbb{R} \rightarrow \mathbb{R} \text{ by } h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is

[JEE (Advanced) 2014]

MDY063

4. Let $a, b \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$. Then f is -
 (A) differentiable at $x = 0$ if $a = 0$ and $b = 1$
 (B) differentiable at $x = 1$ if $a = 1$ and $b = 0$
 (C) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$
 (D) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$

[JEE (Advanced) 2016]

MDY064

5. Let $f: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be function defined by $f(x) = [x^2 - 3]$ and $g(x) = |x|f(x) + |4x - 7|f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$

(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$

(D) g is NOT differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

[JEE (Advanced) 2016]

MDY065

6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x + y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbb{R}$. Then, then value of $\log_e(f(4))$ is ____.

[JEE (Advanced) 2018]

MDY066

7. Let $f_1: \mathbb{R} \rightarrow \mathbb{R}$, $f_2: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f_3: \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow \mathbb{R}$ and $f_4: \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$

(ii) $f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$, where the inverse trigonometric function $\tan^{-1} x$

assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

(iii) $f_3(x) = [\sin(\log_e(x + 2))]$, where for $t \in \mathbb{R}$, $[t]$ denotes the greatest integer less than or equal to t ,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

List-I

P. The function f_1 is

Q. The function f_2 is

R. The function f_3 is

S. The function f_4 is

List-II

1. NOT continuous at $x = 0$

2. continuous at $x = 0$ and NOT differentiable at $x = 0$

3. differentiable at $x = 0$ and its derivative is NOT continuous at $x = 0$

4. differentiable at $x = 0$ and its derivative is continuous at $x = 0$

[JEE (Advanced) 2018]

The correct option is :

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

(B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

MDY067

8. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x-1)\sin x$ and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg: \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE ? [JEE (Advanced) 2020]

- (A) If g is continuous at $x = 1$, then fg is differentiable at $x = 1$
 (B) If fg is differentiable at $x = 1$, then g is continuous at $x = 1$
 (C) If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$
 (D) If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$

MDY068

9. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be functions satisfying $f(x+y) = f(x) + f(y) + f(x)f(y)$ and $f(x) = xg(x)$ for all $x, y \in \mathbb{R}$. If $\lim_{x \rightarrow 0} g(x) = 1$, then which of the following statements is/are TRUE? [JEE (Advanced) 2020]

- (A) f is differentiable at every $x \in \mathbb{R}$
 (B) If $g(0) = 1$, then g is differentiable at every $x \in \mathbb{R}$
 (C) The derivative $f'(1)$ is equal to 1
 (D) The derivative $f'(0)$ is equal to 1

MDY069

10. Let the functions $f: (-1, 1) \rightarrow \mathbb{R}$ and $g: (-1, 1) \rightarrow (-1, 1)$ be defined by $f(x) = |2x - 1| + |2x + 1|$ and $g(x) = x - [x]$, where $[x]$ denotes the greatest integer less than or equal to x . Let $fog: (-1, 1) \rightarrow \mathbb{R}$ be the composite function defined by $(fog)(x) = f(g(x))$. Suppose c is the number of points in the interval $(-1, 1)$ at which fog is **NOT** continuous, and suppose d is the number of points in the interval $(-1, 1)$ at which fog is **NOT** differentiable. Then the value of $c + d$ is ____ [JEE (Advanced) 2020]

MDY070

11. Let $f: (0, 1) \rightarrow \mathbb{R}$ be the function defined as $f(x) = [4x] \left(x - \frac{1}{4} \right)^2 \left(x - \frac{1}{2} \right)$, where $[x]$ denotes the greatest integer less than or equal to x . Then which of the following statements is(are) true? [JEE (Advanced) 2023]

- (A) The function f is discontinuous exactly at one point in $(0, 1)$
 (B) There is exactly one point in $(0, 1)$ at which the function f is continuous but **NOT** differentiable
 (C) The function f is **NOT** differentiable at more than three points in $(0, 1)$
 (D) The minimum value of the function f is $-\frac{1}{512}$

MDY071

12. Let $S = (0, 1) \cup (1, 2) \cup (3, 4)$ and $T = \{0, 1, 2, 3\}$. Then which of the following statements is(are) true ? [JEE (Advanced) 2023]

- (A) There are infinitely many functions from S to T
 (B) There are infinitely many strictly increasing functions from S to T
 (C) The number of continuous functions from S to T is at most 120
 (D) Every continuous function from S to T is differentiable

MDY072

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	C	D	A	A	D	D	D	A
Que.	11	12	13	14						
Ans.	B	D	D	A						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C	B,C	A,B,C	B,C	A,B,C,D	A,B,C,D	B,C,D	A,B,D	B,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	C,D	A,C,D	A,B	A,C	C	B	A,C	A	C	

EXERCISE (S)

1.	1	2.	2	3.	8	4.	5150	5.	1
6.	2	7.	4	8.	120	9.	3	10.	10

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	4	3	10	1	1	5	1	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	1	3	79	2	4	4	2	1
Que.	21									
Ans.	15									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	3	A,B	B,C	2	D	A,C	A,B,D	4
Que.	11	12								
Ans.	A,B	A,C,D								